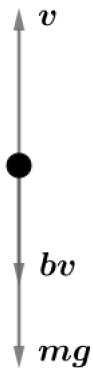


2018A F=ma Exam: Problem 10

Kevin S. Huang



For an object thrown upward with air resistance, we have

$$m \frac{dv}{dt} = -bv - mg$$

$$a = -\frac{b}{m}v - g$$

The object will accelerate downwards to terminal velocity

$$0 = -\frac{b}{m}v_t - g$$

$$v_t = -\frac{mg}{b}$$

while the magnitude of acceleration decreases to 0. Thus, as $t \rightarrow \infty$, $a \rightarrow 0$ so the answer is E.